

would create a layer of abstraction and allow programmers to choose the right tool for each task and allow them to function independently on the same system."

Use of electronic design automation. The software programming mechanism varies widely when moving from single-core to multicore. Vasudevan shares, "Design engineers need to know proper usage of electronic design automation (EDA) tools involved in the design to enable the multicore system run maximum number of functions in parallel. EDA tools play an important role in analysing the application, figuring out functions that can be automatically run in parallel, etc."

Applications based on multicore

One of the fastest methods to bundle more features within the tight budgets is to add additional task-specialised silicon either as heterogeneous cores or chips from third-party suppliers that can be put to sleep or turned off entirely when not required. As a result, there is great scope for multicores.

Some of the applications where multicores make a significant mark include consumer electronics (tablets, smartphones, gaming devices, etc), automotive, networking and communication, and aerospace and defence.

The pressure to pack more features (like wireless connectivity, video/flash/audio playback and video editing/creation/conferencing) into devices with smaller form factors—in order to create product differentiation—and also release the product faster in the market is driving smartphone manufacturers towards multicore.

Applications that use complex algorithms with advanced signal processing now depend on multicore processors to produce smarter devices. Panikkar cites, "Multicore is also of great use in safety-critical systems, where creating redundancy and dedicating cores to specific time-

critical tasks can maximise safety and reliability."

"Alongside, one is witnessing an increasing demand from tablets and gaming devices. Multicore has been around for a long time but the recent tablet revolution (with the recent Tegra-2 and Snapdragon) has brought it to the mainstream," adds Vasudevan.

Listing the benefits of multicore in certain applications, Satyan Namdhari, an embedded professional and idea tinker, shares, "Devices as small as a hearing aid use multicore processing to decrease power consumption, as power consumption increases proportionally with frequency. Mobile phones too use multicore for this and other reasons like specialised processing by each core. For telecom infrastructure, one of the performance parameters is the number of channels processed per PCB or equipment. Multicore chips are used here to increase this type of performance."

Sharing his experience in automotive design, Ravi Shankar Kashi, consultant at Tata Consultancy Services, says, "A loaded automotive infotainment system has multiple inputs (voice commands, touch-screen, key panels, encoders and GPS), multiple communication buses and multiple output devices (graphical LCD dashboards and speakers), file systems (mass storage devices ranging from hard disks to USB pen drives), wired communication (CAN, LIN, MOST and Ethernet), wireless connectivity (WiFi and GPRS) and lot many other features and capabilities. Here ARM+DSP or MIPS+DCP SoC combination has been used so far. SoCs with asymmetric cores like DaVinci (ARM+DSP), however, will be more relevant in automotive and communication areas."

Pathak informs that automotive OEMs have already started using electronic engine control units based on multicore.

W. Peter Seng, software engineer at Continental Automotive Systems, adds that the first goal for multicore

is obviously more performance. But with a large increase in the complexity of embedded systems, especially in automobiles, the other main goal for multicore is reliability (using 'lock-step' processors) to meet the requirements for standards such as ISO26262; there is no improvement in the performance in terms of instructions per second, but the system as a whole is more reliable.

Thronson shares, "Commercial networking applications have embraced multicore and multithreading to a great extent in both control plane and data plane applications. In fact, some networking applications are moving to many-core implementations that incorporate larger numbers of cores."

Multicore is here to stay and grow

Although multicore sounds fancy to many embedded designers, it's evident that multicore will soon become a norm in many embedded applications. Use of multicore is increasing in the embedded market but the adoption rate varies from one market segment to another. Currently used in complex embedded applications, multicore chips will eventually become a part of low-power, battery-powered embedded devices.

Kumaran says, "Product development is likely to be more complicated in a multicore environment since true parallelism requires different programming techniques than in single-core systems." However, he adds, the benefits outweigh the increase in complexity and development costs.

Multicore software development emphasises the need for improved tools and system analysis. But Vasudevan strongly believes that the multicore ecosystem—processors, EDA tools, engineering skills—will evolve as devices like tablets start reaching consumers. Given its benefits, multicore is truly here to stay and grow! ●

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